

DATA ANALYTICS PROJECT REPORT



Uber Power BI Dashboard

[Bookings](#) | [Revenue](#) | [Vehicles](#) | [Customers](#) | [Locations](#)

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Tool: Microsoft Power BI | Dataset: 150,000 ride records

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1. Executive Summary

This report documents an end-to-end **Power BI analytics dashboard** built on Uber ride data. The dashboard turns roughly **150,000 ride-level records** into insights about **bookings, revenue, vehicles, customers and locations**, so that operational and strategic decisions can rest on evidence rather than intuition.

The solution is organised as a landing page plus five analytical pages (Overview, Vehicle, Revenue, Customer and Location), linked by interactive navigation buttons and filters. Each page answers a specific business requirement, from executive snapshots to revenue-leakage detection and peak-demand analysis.

Item	Detail
Project	Uber Power BI Dashboard (learning, portfolio and demonstration project)
Author	Lakshay Sharma, Aspiring Data Analyst
Tool stack	Microsoft Power BI, Power BI Service / Microsoft Fabric, DAX, data modeling
Data	Sample ride-level dataset: about 1,50,000 rows (150,000) and more than 10 features
Analysis grain	Monthly and quarterly trends
Structure	Landing page + 5 analytical pages with interactive filters and navigation

Headline capabilities

- Monitor total bookings, lost bookings, revenue, total distance and average distance per ride at a glance.
- Compare vehicle types on revenue contribution and on ride completion versus incompleton.
- Track revenue by month, quarter, vehicle, payment method and customer, and estimate lost revenue.
- Segment customers as first-time, returning or regular, and quantify the revenue impact of cancellations.
- Find top pickup and drop areas and the peak demand time slots through a day-by-time heatmap.

2. Problem Statement and Objectives

Uber operates at very large scale, generating thousands of rides every day. Raw ride data on its own cannot guide decisions; it has to be shaped into meaningful, comparable measures. The project targets three recurring challenges:

Challenge	Why it matters
Revenue loss	Incomplete or cancelled rides remove revenue that was otherwise expected, and leakage is hard to see in raw rows.
Cancellations	Customer cancellations, and the reasons behind them, directly hurt revenue and customer experience.
Demand patterns	Without knowing where and when demand peaks, driver allocation and pricing are less efficient.

Business objectives

#	Objective
1	Monitor overall ride and revenue performance
2	Identify revenue drivers and loss areas
3	Analyze vehicle-wise contribution and efficiency
4	Understand customer behavior and the impact of cancellations
5	Identify peak-demand locations and time slots
6	Enable data-driven operational and strategic decisions

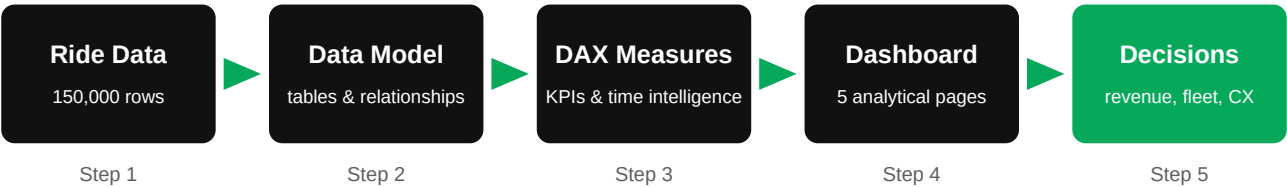
3. Dataset Overview

The dataset holds **ride-level transactional data**: one record per booking. It is a sample dataset of about 150,000 rows with more than ten features, analysed at monthly and quarterly levels to expose trends and patterns.

Attribute group	Examples of fields	Used for
Booking details	Booking ID, status, booking value	Bookings, lost bookings, revenue, completion rates
Vehicle types	Auto, Bike, Sedan, XL and others	Fleet contribution and efficiency
Customer information	Customer identity and history	Segmentation, top customers, cancellation behavior
Locations	Pickup and drop locations	Top areas, demand hotspots
Distance	Distance traveled per ride	Total and average distance, revenue per km
Time and date	Date, month, quarter, time slot, day	Trends, peak slots, heatmap
Ratings and cancellations	Customer and driver ratings, cancellation reasons	Service quality, top cancellation reason
Payment	UPI, Cash, Wallet, Cards	Revenue by payment method

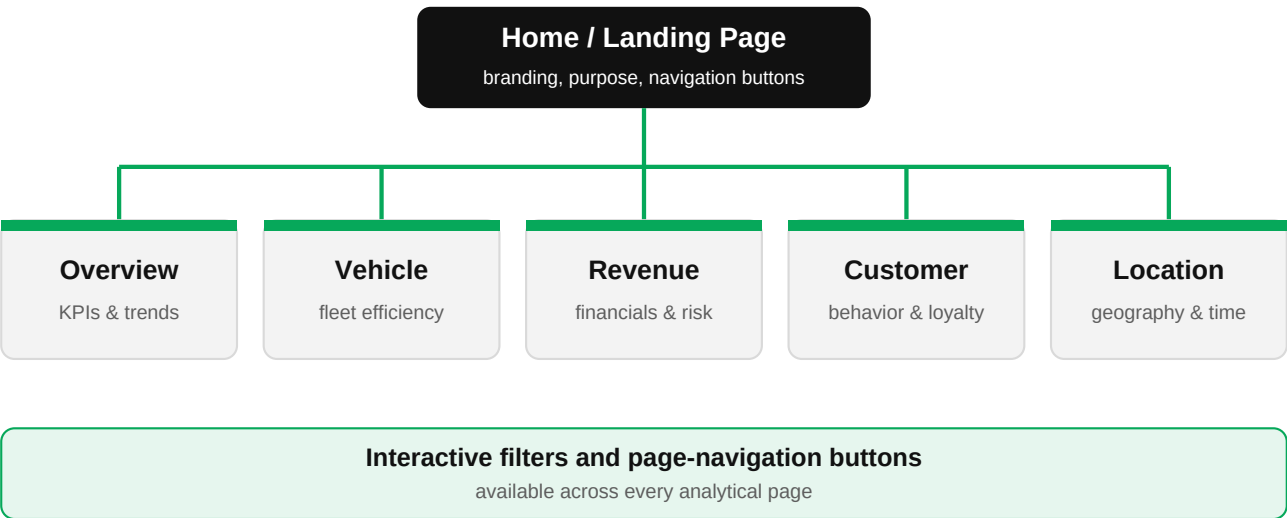
4. Solution Approach and Architecture

The project follows the standard analytics pipeline: raw data is modeled, enriched with DAX measures, presented through purpose-built pages, and finally used to inform decisions.



Dashboard structure

A landing page introduces the dashboard and routes users to five analytical pages. Filters and navigation buttons let users move between pages without losing context.



How each page serves the business objectives

Objective	Overview	Vehicle	Revenue	Customer	Location
Monitor overall performance	●				
Identify revenue drivers and losses	●		●		
Vehicle contribution and efficiency		●			
Customer behavior and cancellations				●	
Peak demand locations and time slots	●				●

Mapping is the report author's reading of the page descriptions; pages often support several objectives.

5. Page-by-Page Analysis

5.1 Home / Landing Page

Purpose. Introduces the Uber analytics dashboard, gives context, and provides navigation to every analytical page.

- Uber branding and visual identity
- A brief description of the dashboard's purpose
- Navigation buttons to all analytical pages

BUSINESS VALUE

Improves user experience and makes the dashboard stakeholder-ready and portfolio-ready.

5.2 Overview Page

Business requirement. Provide a high-level snapshot of Uber's operational and financial performance.

KPIs displayed

Total Bookings	Lost Bookings	Total Revenue	Total Distance	Avg Distance per Ride
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Insights provided

- Monthly and quarterly booking trends
- Revenue trends over time
- Revenue by vehicle type
- Top pickup and drop locations
- Average customer and driver ratings

BUSINESS VALUE

Enables quick, executive-level decision-making and highlights overall growth, decline or inefficiency.

5.3 Vehicle Page

Business requirement. Analyze performance at the vehicle level to optimize fleet usage.

Key metrics

Booking count by vehicle	Revenue by vehicle type	Revenue contribution %	Ride completion rate	Ride incomplete rate
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Insights provided

- Identification of the top revenue-generating vehicles
- Comparison of completion efficiency across vehicle types
- Sparkline trends for completed bookings

BUSINESS VALUE

Supports fleet optimization and helps improve pricing and incentive strategies.

5.4 Revenue Page

Business requirement. Provide detailed financial insights and identify revenue risks.

Key analysis

- Monthly and quarterly revenue trends
- Revenue by vehicle type
- Revenue by payment method (UPI, Cash, Wallet, Cards)
- Revenue by top customers

Efficiency and risk metrics

Month-on-Month revenue change	Average revenue per booking	Revenue per kilometer	Lost revenue estimation
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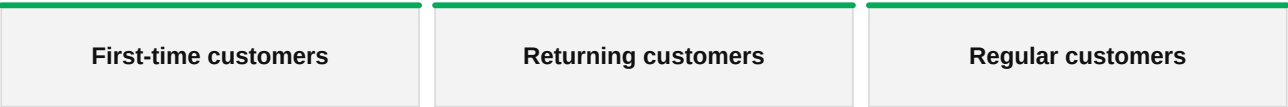
BUSINESS VALUE

Identifies profitable segments, detects revenue leakage, and supports financial planning and strategy.

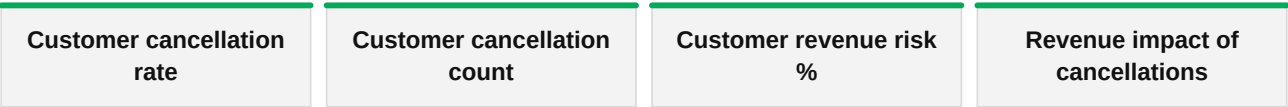
5.5 Customer Page

Business requirement. Understand customer behavior, loyalty and the impact of cancellations.

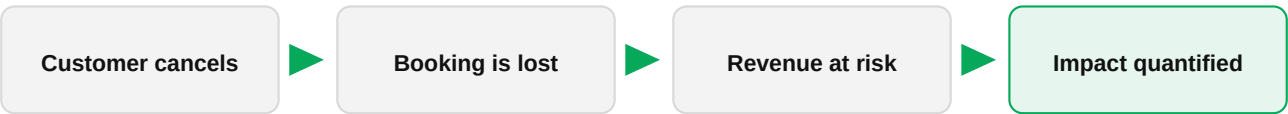
Customer segmentation



Key metrics



From cancellation to revenue impact



Insights provided

- Top customer cancellation reason (for example, Wrong Address)
- Customer trend over time
- Payment method preference
- A detailed customer-level table

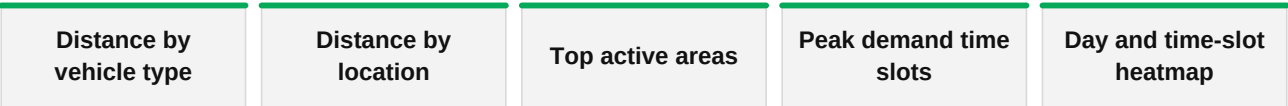
BUSINESS VALUE

Improves retention strategy, reduces revenue lost to cancellations and enhances customer experience.

5.6 Location Page

Business requirement. Analyze geographic and time-based demand patterns.

Key insights



- **Total distance by vehicle type** shows how each fleet category contributes to kilometers covered.
- **Distance covered by location and top active areas** reveal where operations concentrate.
- **Peak time slots and the day-wise heatmap** show when demand is highest.

BUSINESS VALUE

Optimizes driver allocation, supports surge-pricing decisions and improves city-level operations.

6. Key Metrics Glossary

Plain-language meaning of the main measures used across the dashboard. Formulas describe the usual definition of each measure; exact DAX expressions live in the Power BI file.

Metric	Page	Meaning
Total Bookings	Overview	Count of ride bookings in the selected period
Lost Bookings	Overview	Bookings that did not result in a completed ride
Average Distance per Ride	Overview	Total distance divided by number of rides
Revenue Contribution %	Vehicle	A vehicle type's revenue as a share of total revenue
Completion / Incomplete Rate	Vehicle	Share of bookings completed, or not completed, per vehicle type
Month-on-Month Change	Revenue	Percentage change in revenue versus the previous month
Average Revenue per Booking	Revenue	Total revenue divided by number of bookings
Revenue per Kilometer	Revenue	Total revenue divided by total distance traveled
Lost Revenue (estimate)	Revenue	Estimated value of revenue lost through non-completed bookings
Customer Cancellation Rate	Customer	Share of bookings cancelled by customers
Customer Revenue Risk %	Customer	Portion of revenue exposed to customer cancellations

7. Tools and Technical Skills

Tool / Skill	Role in the project
Microsoft Power BI	Report design, visuals, interactivity, navigation
Power BI Service / Microsoft Fabric	Publishing and hosting the live, shareable report
DAX (Data Analysis Expressions)	Calculated measures for KPIs, rates, ratios and risk estimates
Data modeling and relationships	Structuring ride data so filters flow correctly across pages
Time intelligence	Monthly and quarterly trends, month-on-month comparison
KPI design and dashboard UX	Clear hierarchy, consistent layout, easy navigation

8. Business Impact

The dashboard equips Uber stakeholders to act on data rather than assumption:

Impact area	How the dashboard helps
Performance tracking	Track business performance through KPIs and trends in real time (once live data is connected)
Revenue optimization	Identify revenue growth and loss areas, including payment method and top-customer views
Fleet and driver utilization	Compare vehicle types and align drivers with peak locations and time slots
Cancellation reduction	Surface top cancellation reasons and their revenue effect
Customer satisfaction	Understand loyalty segments and ratings to improve experience
Decision quality	Give leaders a shared, data-driven view for operational and strategic choices

9. Key Learnings and Future Enhancements

Key learnings

- End-to-end Power BI dashboard development
- Translating business requirements into analytics
- Practical use of DAX for real-world problems
- Data storytelling and professional dashboard design
- UX considerations for enterprise dashboards

Future enhancements

Enhancement	Expected benefit
Real-time data integration	Live monitoring instead of periodic snapshots
Predictive demand forecasting	Plan driver supply and pricing ahead of demand peaks
Customer churn prediction	Intervene early with customers likely to leave
Driver performance analytics	Extend efficiency analysis from vehicles to drivers
Advanced AI-based insights	Automated detection of patterns and anomalies

10. Conclusion

The Uber Power BI Dashboard shows how raw ride data can be transformed into a clear, decision-ready story. By organising the analysis into Overview, Vehicle, Revenue, Customer and Location pages, it covers the full cycle from monitoring performance to diagnosing revenue loss, understanding cancellations, and planning around demand.

Beyond the dashboard itself, the project demonstrates practical skills that matter for a data analyst: data modeling, DAX, KPI design and business storytelling. The planned extensions, real-time data and predictive analytics, would move the solution from describing what happened to anticipating what comes next.

Appendix: Project Links and Author

Item	Detail
Live dashboard	Open the Power BI report
GitHub repository	github.com/User0406/uber-Analysis
Author	Lakshay Sharma, Aspiring Data Analyst Power BI Data Analytics
Note	Created for learning, portfolio and demonstration purposes using a sample dataset.